

Amar Dhillon

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EDUCATION

University of Maryland, College Park
B.S. in Computer Science, Minor in Statistics; GPA: 3.5

College Park, MD
Expected May 2028

CERTIFICATIONS

Amazon Web Services (AWS) Certified Developer - Associate
Amazon Web Services

Nov 2024 – Nov 2027
Digital Badge

PROJECTS

- IEX Order Book & Matching Engine** | *C++17, STL, POSIX, Linux perf* Jul 2026 – Aug 2026
- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
 - Implemented B+ tree and LSM-tree backends for the event log and benchmarked them on identical 2M-record workloads, where the B+ tree gave **12.8x** the read throughput and 22x faster range queries at the cost of 27% more disk space
 - Traced an apparent 1.65x slowdown to the CPU frequency governor using perf counters and a null control book, then cut run-to-run variance from about 6% to 0.06% with core pinning and a fixed clock
 - Stored price levels in a `std::map` of `std::list` FIFOs with a hash map from order ID to list iterator for $O(1)$ cancels, sustaining **1.26M events/sec** over the full replay and a 1.2 μ s per-update p99 on a separate 2M-event run

EXPERIENCE

- Software Development Engineer Intern** Sep 2026 – Dec 2026
Amazon Web Services Redmond, WA
- Writing Java logic that sets the upper and lower bounds on EC2 capacity Amazon Redshift reserves for operations like patching, balancing availability against cost, and backtesting it

- Software Engineering Intern** Jun 2026 – Aug 2026
Capital One Richmond, VA
- Cut AI workflow generation time **65%** to under 15s by replacing a Claude skill's open-ended search with focused sub-skills that call Python tools for workflow search and validation, guided by CloudWatch and OpenSearch latency logs
 - Because the 270K+ row PostgreSQL workflow table had strict access limits, precomputed popular workflows into a category tree that the Claude skill searches first, with Redis caching full workflows to cut database queries

- Software Engineer (Contract)** Jan 2026 – Present
UMD App Development Contracting (Client: Amazon Leo) College Park, MD
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

- Machine Learning Research Intern** Dec 2025 – Present
UMD Institute for Advanced Computer Studies (UMIACS) College Park, MD
- For a PhD-led robotics project aimed at ICLR, built a pipeline that shows a vision-language model AnyGrasp's candidate grasps as Open3D overlays on multi-camera images, then tuned prompts so the model picks the right grasp more often

- Co-Founder, Software Engineer** Jun 2025 – Present
TerpLabs College Park, MD
- Co-founded a 110+ member student product studio that has shipped 5 live products
 - Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with **20,000+ visits** in its first 30 days, to reduce database load and latency

TECHNICAL SKILLS

Languages: C++17, C, Python, Java, Rust, SQL, TypeScript, Bash
Systems & Performance: low-latency C++, debugging and profiling with Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, data structures and algorithms, benchmarking
Cloud & Tools: AWS (Lambda, Kinesis, EC2, Fargate, CloudWatch monitoring), AWS CDK, PostgreSQL, Redis, Docker, GitHub Actions (CI/CD), Git, Pytest